

UNDERSTANDING RADIOACTIVITY

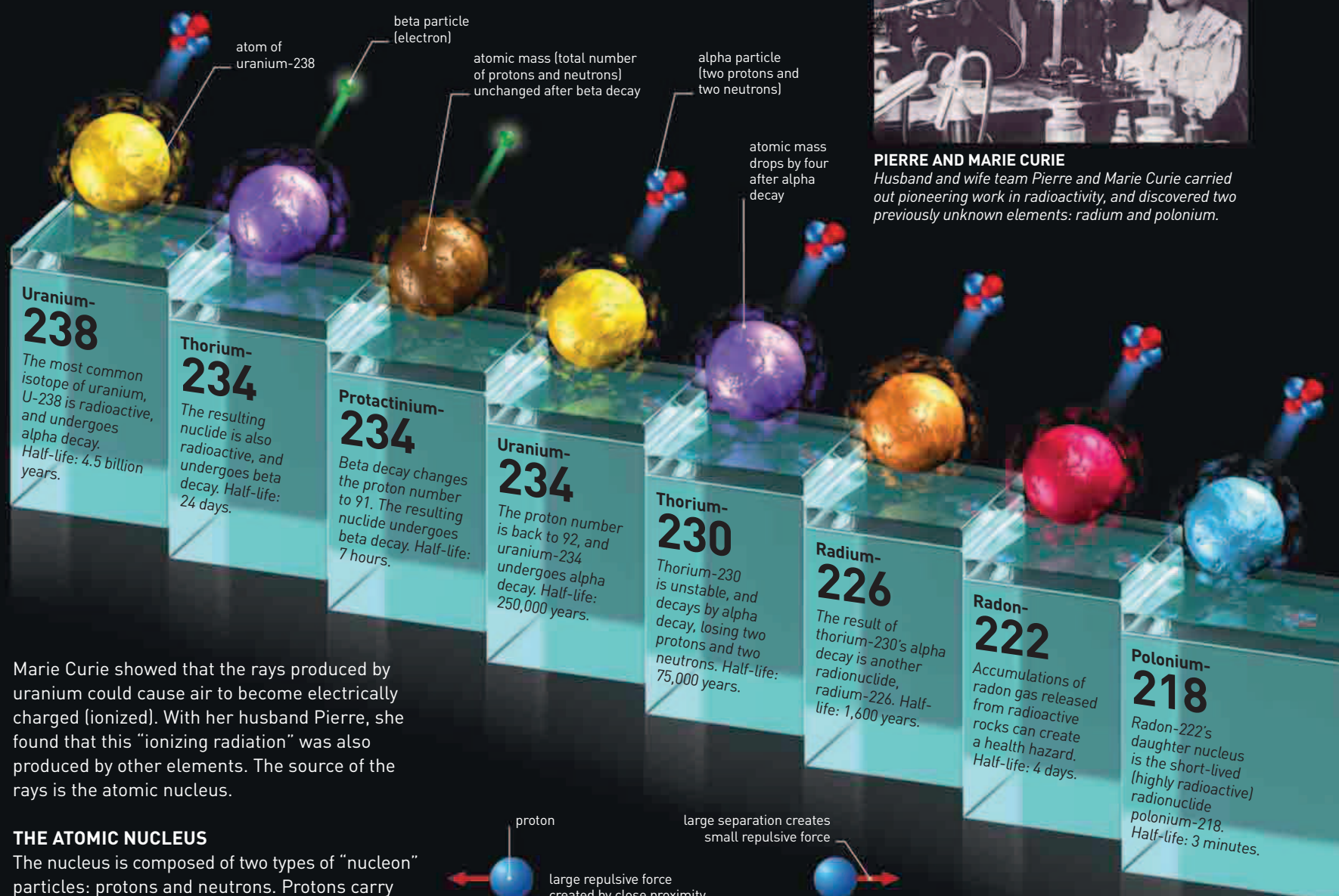
THE DISCOVERY THAT SOME ELEMENTS ARE RADIOACTIVE TRIGGERED A REVOLUTION IN PHYSICS

In 1896, French physicist Henri Becquerel found that compounds containing uranium produce invisible radiation. Within months, Polish scientist Marie Curie showed that the rays emanate from the uranium atoms themselves, and in 1898 she coined a term to describe the phenomenon: radioactivity.



PIERRE AND MARIE CURIE

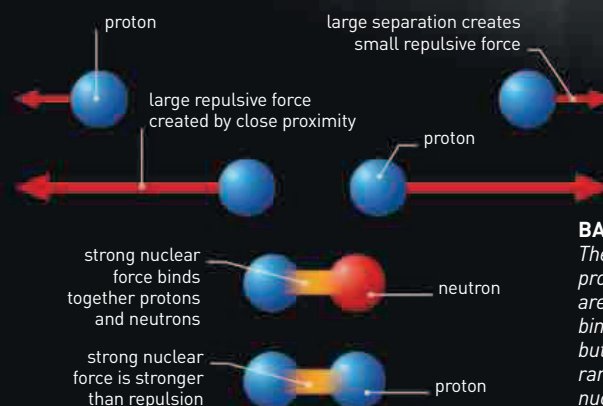
Husband and wife team Pierre and Marie Curie carried out pioneering work in radioactivity, and discovered two previously unknown elements: radium and polonium.



Marie Curie showed that the rays produced by uranium could cause air to become electrically charged (ionized). With her husband Pierre, she found that this "ionizing radiation" was also produced by other elements. The source of the rays is the atomic nucleus.

THE ATOMIC NUCLEUS

The nucleus is composed of two types of "nucleon" particles: protons and neutrons. Protons carry positive charge, so they repel each other, but the "strong nuclear force" binds the nucleons together. The number of protons in the nucleus indicates the element to which an atom belongs. There are different versions (isotopes) of each element, which differ in their number of neutrons. A particular combination of protons and neutrons is called a nuclide—or, if unstable, a radionuclide.



BALANCE OF FORCES

The repulsive (electrostatic) force between protons is stronger the closer the protons are to each other. The strong nuclear force binds protons and neutrons tightly together, but it only acts over an incredibly short range. This is the main reason why larger nuclei tend to be unstable.